
Task Arithmetic for Efficient Machine Unlearning: A Study on CNN Dynamics

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Abstract

This work evaluates *Task Arithmetic* for single-class unlearning across VGG-11, ResNet-18, and MobileNetV2. While state-of-the-art metrics (ZRF, KL Divergence, MIA) validate apparent knowledge erasure, we expose critical post-ablation topological vulnerabilities: *Semantic Shift* and *Over-forgetting*. Finally, the **Anamnesis Index (AIN)** quantifies latent residual traces and the resulting “Rebound Effect,” revealing that true permanent amnesia remains an illusion.

1. Introduction

The “Right to be Forgotten,” formally codified in regulations such as the GDPR (European Parliament and Council, 2016), has made it essential to develop techniques capable of selectively erasing specific information from trained neural networks. While the gold standard remains retraining from scratch on the retain dataset, the computational cost is often prohibitive. Machine Unlearning aims to bridge this gap by providing efficient alternatives for data removal.

In this project, the efficacy of **Task Arithmetic** is explored as a lightweight framework for selective unlearning in Convolutional Neural Networks (CNNs). The framework is evaluated on the CIFAR-10 dataset across three architectures with distinct structural properties: VGG-11 (purely sequential), ResNet-18 (residual connections), and MobileNetV2 (bottleneck efficiency). This diversity allows for a critical analysis of how model topology affects the stability and quality of the unlearning process.

The main contribution of this work is an evaluation of CNN vulnerabilities during vectorial weight manipulation. Knowledge erasure is rigorously validated through state-of-the-art cryptographic metrics, such as Kullback-Leibler (KL) Divergence, Zero Retrain Forgetting (ZRF) Score,

and Membership Inference Attacks (MIA), while critical topological side-effects, specifically *Semantic Shift* and *Over-forgetting*, are exposed. Furthermore, the Anamnesis Index (AIN) (Chundawat et al., 2023b) is utilized to quantify latent residual traces and the resulting “Rebound Effect” during localized retraining. Finally, Task Arithmetic is calculated to operate at a fraction of the computational cost of native retraining, validating the framework as a scalable solution for real-world privacy compliance.

2. Related Work

Machine Unlearning and Weight Manipulation. Traditional unlearning approaches often require prohibitive retraining pipelines. Recently, *Task Arithmetic* (Ilharco et al., 2023) introduced direct weight-space manipulation via “Task Vectors” computed as the difference between fine-tuned and pre-trained weights. Originally proposed to alter behaviors in large foundation models across both text and vision domains, this framework is specifically adapted here for localized semantic ablation. While prior works leveraged weight manipulation primarily for compression (e.g., deep pruning (Han et al., 2015)), this study shifts the focus toward privacy compliance.

Architecture and Optimization. Existing literature highlights that the inductive biases of CNN topologies, such as sequential density (Simonyan & Zisserman, 2015), residual routing (He et al., 2015), and compressed bottlenecks (Sandler et al., 2019), dictate latent space geometry. Furthermore, optimization strategies heavily impact parameter stability: standard adaptive optimizers couple weight decay with gradient updates, leading to unregularized parameter growth, whereas decoupled optimizers preserve the structural constraints of the weight space (Loshchilov & Hutter, 2019).

3. Methodology

3.1. Architectural Framework and Training Plasticity

To evaluate structural dependencies in Machine Unlearning, three diverse architectures are analyzed: VGG-11 with Batch Normalization (sequential) (Simonyan & Zisserman,

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2015; Ioffe & Szegedy, 2015), ResNet-18 (residual) (He et al., 2015), and MobileNetV2 (bottleneck) (Sandler et al., 2019). All models are trained from scratch on CIFAR-10, incorporating standard topological adaptations—such as replacing ResNet-18’s initial 7×7 downsampling with a 3×3 convolution—to accommodate the dataset’s 32×32 resolution.

A fundamental design choice is the use of the **AdamW optimizer** (Loshchilov & Hutter, 2019). By decoupling weight decay from gradient updates, AdamW prevents the uncontrolled growth of weight magnitudes often observed in standard optimizers. This structural regularization is critical for Task Arithmetic: by maintaining a constrained and stable weight space, it ensures that the derived task vectors isolate genuine semantic shifts rather than unregularized magnitude artifacts, thereby maximizing the precision of the vectorial subtraction.

3.2. Expert Generation via Directional Shift

The unlearning procedure requires a task-specific vector τ . To obtain this, an “Expert Model” is generated by fine-tuning the pre-trained base model exclusively on the target class (Class 8: Ship), chosen for its relative semantic isolation from animal classes to minimize noise from shared visual primitives. To prevent the expert from losing global feature knowledge—which would corrupt the subsequent task vector, a **Directional Shift** strategy is implemented: the fine-tuning is strictly bounded to 5 epochs with a reduced learning rate (1×10^{-4}). This constrained optimization ensures that the resulting vector $\tau = \theta_{ft} - \theta_{pre}$ represents a clean, targeted gradient toward the specific concept without destroying the underlying convolutional filters.

3.3. Surgical Ablation and Hyperparameter Tuning

The surgical intervention on the base model weights θ_{pre} is formalized as:

$$\theta_{unl} = \theta_{pre} + \alpha \cdot \text{mask}(\tau, \rho) \quad (1)$$

where α is the scaling factor (Ilharco et al., 2023) and ρ is the drop-percentile for magnitude-based pruning (Han et al., 2015). To optimize (α, ρ) , a comprehensive **Grid Search** was executed exploring $\alpha \in \{-0.5, -1.0, \dots, -3.0\}$ and $\rho \in [0.50, 0.99]$ (20 linear steps).

The final optimal parameters and their corresponding unlearning performance on the anchor class (Ship) are summarized in Table 1. To contextualize the degradation, these metrics are compared against the base models’ original overall accuracy. Notably, VGG-11 achieved perfect erasure (target accuracy plummeting from 93.5% to 0.0%) while preserving a high retain accuracy (85.5%, experiencing a minimal drop from its original 87.4% per-

formance). This confirms that sequential architectures allow for high-magnitude interventions ($\alpha = -2.5$) when applied to a highly sparse selection of weights ($\rho = 0.990$). In contrast, ResNet-18 and MobileNetV2 required more conservative interventions to achieve erasure (target accuracy dropping from 95.0% to 3.0% and from 87.2% to 0.1%, respectively), highlighting the extreme sensitivity of skip-connections and compressed bottlenecks to weight perturbations, which ultimately forces a severe degradation of the retain set (62.3% and 57.1%).

Table 1. Optimal hyperparameters (α, ρ) derived from the Grid Search, and performance comparison against the original base model accuracy.

Architecture	α	ρ (Drop)	Orig. Acc.	Retain Acc.	Forget Acc.
VGG-11 (bn)	-2.5	0.990	87.4%	85.5%	0.0%
ResNet-18	-1.0	0.629	89.1%	62.3%	3.0%
MobileNetV2	-1.5	0.500	77.1%	57.1%	0.1%

4. Experiments and Results

4.1. Performance Floor and Baselines

To establish a rigorous performance floor, Task Arithmetic (TA) is benchmarked against two standard optimization-based unlearning techniques: *Gradient Ascent* (GA) and *Random Labeling* (RL). As shown in Table 2, optimization-based baselines prove either highly destructive or ineffective depending on model topology. In the purely sequential VGG-11, GA indiscriminately maximizes the loss, inducing a catastrophic collapse of the global feature space (retain accuracy plummets to 11.1%). Conversely, RL applied to ResNet-18 protects the retain set (84.9%) but fails to unlearn the target concept, leaving a residual forget accuracy of 15.9% (indicating simple noise memorization rather than boundary destruction).

TA overcomes these limitations through targeted weight-space surgery. By applying magnitude-based vector subtraction, TA achieves perfect erasure (0.0%) in VGG-11 while preserving 85.5% of the retain set. Notably, MobileNetV2 exhibits generalized degradation across all three methods. This reinforces the hypothesis that highly compressed architectures with linear bottlenecks lack the parameter redundancy required to absorb localized perturbations without structural damage.

4.2. Cryptographic Validation and Empirical Privacy

To assess whether the models genuinely eradicated the target data, their latent distributions are evaluated using **KL Divergence** (distance from a retrained baseline) and **ZRF Score** (Chundawat et al., 2023a) (distance from a randomly initialized network).

Table 2. Comparison: Task Arithmetic (TA) vs Optimization-based Baselines.

Architecture	Task Arithmetic		Grad. Ascent		Rand. Labeling	
	Retain	Forget	Retain	Forget	Retain	Forget
ResNet-18	62.3%	3.0%	46.9%	2.8%	84.9%	15.9%
VGG-11 (bn)	85.5%	0.0%	11.1%	0.0%	47.9%	0.0%
MobileNetV2	57.1%	0.1%	49.6%	4.4%	59.3%	9.6%

As summarized in Table 3, VGG-11 achieves a remarkably low KL divergence and a high ZRF score, indicating its predictive distribution is statistically indistinguishable from native ignorance. Conversely, ResNet-18 and MobileNetV2 exhibit significant KL divergence and notably lower ZRF values. This disparity exposes the phenomenon of “Fake Unlearning”: despite low target accuracy, internal feature correlations remain structured in non-sequential architectures, effectively masking the data rather than purifying it. Finally, **Membership Inference Attacks (MIA)** (Shokri et al., 2017) yield near-random AUCs (~ 0.50) across all models, confirming that target membership is no longer extractable.

Table 3. Privacy Metrics. Optimal unlearning is characterized by low KL divergence, high ZRF score, and MIA AUC near 0.50.

Architecture	KL Div. ($\downarrow$)	ZRF Score ($\uparrow$)	MIA AUC
VGG-11 (bn)	0.97	0.83	~ 0.44
ResNet-18	3.20	0.32	~ 0.43
MobileNetV2	5.91	0.41	~ 0.47

4.3. Topological Side-Effects: Semantic Shift and Over-forgetting

The visual reconfiguration of the latent space provides interesting insights into how different architectures process knowledge ablation, as detailed in the t-SNE and Confusion Matrix visualizations (see Appendix A).

Semantic Shift. Confusion matrices reveal deterministic prediction re-routing rather than random diffusion. In VGG-11, the ablated “Ship” concept shifts predominantly toward “Truck”, indicating reliance on low-level visual primitives. ResNet-18 exhibits a counter-intuitive shift toward “Cat”, suggesting that residual connections bypass suppressed filters and reroute signals to disparate dominant extractors. MobileNetV2 exhibits a chaotic collapse toward ‘Dogs’, indicating its structural fragility.

Over-forgetting. This phenomenon occurs when the ablation of a target concept unintentionally degrades the model’s knowledge of visually similar, untargeted classes. To quantify this collateral damage, the accuracy gap between the proximal “Airplane” class and the remaining safe

classes is measured. VGG-11 demonstrates excellent concept isolation with a negligible 3.4% gap (82.5% Airplane vs 85.9% Others). Conversely, ResNet-18 suffers a severe 13.3% gap (50.5% vs 63.8%), proving that residual connections entangle feature extractors across visually similar classes, hindering surgical ablation. MobileNetV2 similarly registers an 8.9% gap (49.2% vs 58.1%), confirming its architectural sensitivity to localized perturbations.

4.4. The Rebound Effect and Industrial Efficiency

To quantify latent residual traces, the **Anamnesis Index (AIN)** (Chundawat et al., 2023b) is evaluated. Defined as $AIN = E_{unl}/E_{comp}$, it measures the epochs required for the unlearned model to regain target knowledge versus a native comparison model. A value of $AIN < 1.0$ exposes the *Rebound Effect*, indicating that unlearned weights retain a recovery head-start over native ignorance.

VGG-11 exhibits the most severe rebound ($AIN = 0.36$), recovering the target concept in only 4 epochs (vs. 11 baseline). MobileNetV2 similarly bypasses the cold-start barrier ($AIN = 0.53$, converging in 16 vs. 30 epochs), while ResNet-18 shows a more moderate rebound ($AIN = 0.75$, 9 vs. 12 epochs). These results validate the persistence of weight-space artifacts across diverse inductive biases, proving that surgical manipulation leaves latent feature representations that the optimizer can rapidly re-activate.

Green AI and Computational Efficiency. The framework achieves an 88.27% computational reduction, requiring 381.36 seconds for VGG-11 versus 3251.29 for full re-training. While expert fine-tuning incurs a 378.98-second upfront cost, surgical unlearning executes in just 2.39 seconds via $O(1)$ vector subtraction (Ilharco et al., 2023). This ensures rapid privacy compliance while minimizing the carbon footprint of model maintenance.

5. Discussion and Conclusions

Task Arithmetic enables efficient machine unlearning, achieving an 88.27% computational reduction versus re-training. However, efficacy remains strictly topology-bound. Sequential architectures (VGG-11) approximate statistical native ignorance with negligible over-forgetting. Conversely, residual (ResNet-18) and bottleneck (MobileNetV2) topologies resist surgical erasure, inducing “Fake Unlearning”: they mask data rather than purifying the latent space, causing severe semantic entanglement.

Crucially, the Anamnesis Index (AIN) reveals permanent amnesia remains an illusion. Vectorial subtraction leaves latent weight-space paths, triggering a *Rebound Effect* that accelerates concept re-acquisition. Ultimately, while Task Arithmetic offers a GDPR-compliant “Green AI” solution, this efficiency trades off with cryptographic irreversibility.

6. Statement on the use of AI

During the development of this project, Artificial Intelligence (AI) was utilized as a support tool. Its application was confined to the following activities:

- **Literature Review and State of the Art (SOTA):** Assistance in navigating recent scientific literature regarding Machine Unlearning. Specifically, building upon our pre-existing familiarity with the Task Arithmetic framework (derived from a prior NLP project), AI was employed to research and evaluate alternative ablation methodologies and comparative metrics, thereby enriching the theoretical background (e.g., the concept of Zero Retrain Forgetting).
- **Validation and Feasibility Analysis:** Critical brainstorming to assess the temporal and computational feasibility of proposed ideas. This helped filter out excessively demanding tests given the available hardware, favoring more targeted and efficient experimental setups and metrics.
- **Debugging and Low-Level Troubleshooting:** Assistance in diagnosing PyTorch bottlenecks and crashes. Specific interventions included resolving a memory allocation error (Segmentation Fault) related to hardware acceleration on Apple MPS chips, and fixing a Batch Normalization layer crash caused by singleton batches during dynamic dataset filtering.
- **Technical Consultation and Parametric Choices:** Utilizing AI as a sounding board to overcome experimental impasses. Specifically, we consulted the AI to: (1) Evaluate the opportunity to expand the search space for the hyperparameter α , having observed that the VGG-11 architecture was saturating at the established lower bounds (-2.0); (2) Discuss the calibration of the Learning Rate during re-learning tests (Anamnesis Index). The AI validated our hypothesis to drastically lower the LR to prevent instantaneous re-acquisition, enabling precise measurement of latent residual traces.
- **Data Analysis and Visualization:** Assistance in the comparative analysis of the obtained metrics and support in writing the code (e.g., Matplotlib/Seaborn) required to generate appropriate visual representations. The AI facilitated the graphical translation of numerical results, particularly for plotting latent space clusters (t-SNE) and generating side-by-side confusion matrices.
- **Academic Revision and Terminology:** Support during the drafting phase for lexical optimization, ensuring the adoption of a rigorous technical vocabulary aligned with Deep Learning research standards.

Overall, throughout this work, Artificial Intelligence acted exclusively as a technological facilitator and consultation tool.

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Appendix

A. Visual Proofs of Topological Side-Effects

To facilitate reproducibility, the complete source code is publicly available at: <https://github.com/giulapietrangeli/Task-Arithmetic-Machine-Unlearning.git>.

This appendix provides the qualitative visual evidence for the Semantic Shift and latent space reconfiguration discussed in Section 4.

A.1. t-SNE Latent Space Visualizations

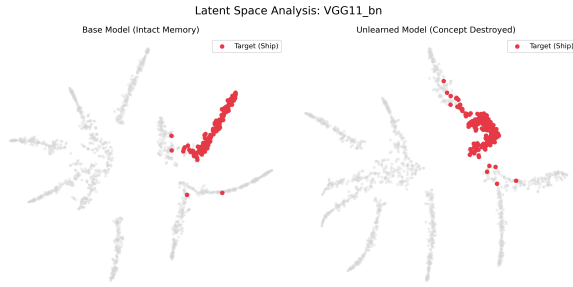


Figure 1. t-SNE latent space visualization for VGG-11 post-unlearning. The red cluster represents the target class (Ship), which is infiltrating nearby different clusters.

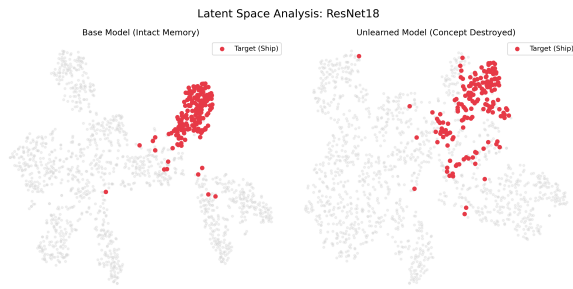


Figure 2. t-SNE latent space visualization for ResNet-18 post-unlearning. The red cluster representing the target class (Ship) is fragmented and infiltrates adjacent distributions.

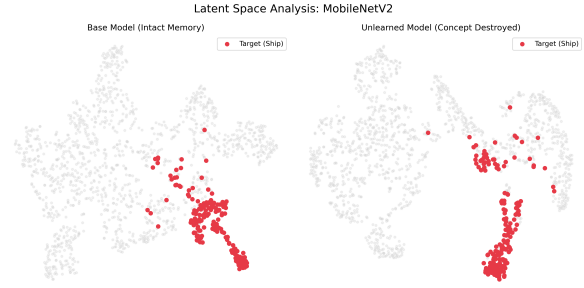


Figure 3. t-SNE latent space visualization for MobileNetV2 post-unlearning. The target class (Ship) remains mostly isolated but is translated to a peripheral region.

A.2. Confusion Matrices

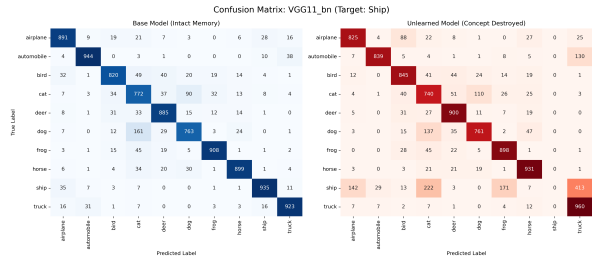


Figure 4. Confusion Matrix for VGG-11 post-unlearning. Note the deterministic semantic shift, with the majority of "Ship" instances misclassified as "Truck".

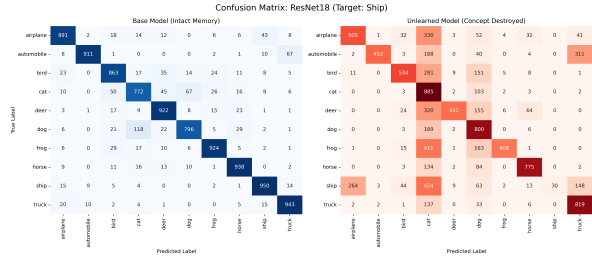


Figure 5. Confusion Matrix for ResNet-18 post-unlearning. The pronounced and unexpected shift of "Ship" predictions toward "Cat" illustrates the complex and unpredictable latent signal generated by residual connections when a target concept is ablated.

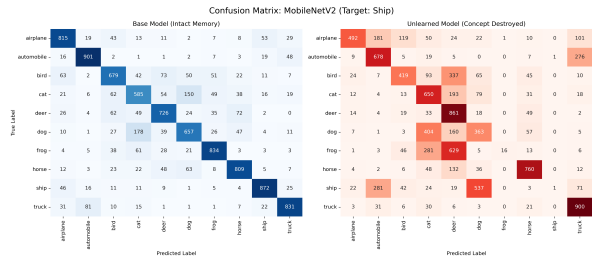


Figure 6. Confusion Matrix for MobileNetV2 post-unlearning. The chaotic redirection of predictions toward unrelated classes like "Dog" visually corroborates the architectural collapse.